COMET: Comic Multimodal Extraction and Translation

Jay Parmar
ja339952@ucf.edu
University of Central Florida
Orlando, Florida, USA

Abstract

This milestone presents COMET, a multimodal pipeline for automatic manga translation and layout-preserving re-typesetting. The system combines image preprocessing, speech-text detection using PaddleOCR, Japanese-to-English machine translation with a MarianMT model, and layout-aware text reinsertion. Unlike standard machine translation systems, COMET preserves visual structure by merging overlapping OCR boxes, removing redundant detections, and fitting translated text within detected regions using heuristic font resizing. Preliminary results on sample manga panels demonstrate successful text extraction, translation, and reintegration while maintaining page readability. We analyze OCR confidence, translation length expansion, and layout overflow rates, and discuss current failure cases including stylized fonts and vertical text. These results demonstrate the feasibility of multimodal document translation requiring joint visual and textual reasoning.

1 Introduction and Problem Statement

Manual manga translation requires text extraction, translation, redrawing, and careful layout reconstruction. Although machine translation systems can translate text strings, they do not account for visual structures such as speech bubble placement, font scaling, or spatial constraints. Consequently, automated manga translation requires joint reasoning across vision and language modalities.

The objective of COMET is to build an end-to-end pipeline that:

(1) Detects text regions in manga panels,
(2) Extracts Japanese text,
(3) Translates it to English,
(4) Reinserts translated text while preserving layout.

This project studies whether existing OCR and translation models can be combined with layout heuristics to produce readable translated pages without manual intervention.

2 Related Work

Optical Character Recognition (OCR) systems such as PaddleOCR [1] have demonstrated strong performance on multilingual text detection and recognition in natural images. Neural machine translation systems such as MarianMT [8] enable high-quality translation across language pairs using pretrained transformer architectures.

Recent multimodal research emphasizes the joint reasoning between vision and language models for document understanding and layout-aware processing [2, 6, 7]. However, most systems focus on document parsing or visual question answering rather than structured re-typesetting tasks [3]. Manga translation presents unique challenges including stylized fonts, vertical Japanese text, and constrained bubble geometry [4, 5]. COMET positions itself as a practical multimodal pipeline that integrates OCR, translation, and layout reconstruction for comic documents.

3 Methodology

COMET is implemented as a modular pipeline following the steps described in our proposal. The pipeline is fully automated and processes each panel independently, enabling systematic error analysis and stage-wise evaluation of detection, translation, and layout performance.

3.1 Preprocessing

Input images are converted to PNG format and normalized. We use grayscale conversion, denoising, and Otsu binarization to improve the robustness of the OCR, implemented as a preprocessing stage within the pipeline.

3.2 OCR and Box Refinement

PaddleOCR is used to detect and extract Japanese text. Since manga often produces fragmented detections, we implement:

- Bounding box merging for spatially close detections,
- Removal of watermark boxes,
- Elimination of overlapping boxes where one box contains more than 90% of another.

Each OCR result is stored with confidence scores and bounding coordinates in JSON-format.

3.3 Translation

The extracted Japanese text is translated using the *susiexyf/marian-finetuned-ja-en* model from HuggingFace. Translation is performed per bounding box. We measure the length expansion ratio to estimate the potential layout overflow.

3.4 Text Removal and Re-Typesetting

Original text regions are whitened using bounding-box interior filling. Then, the translated text is rendered back using heuristic layout fitting:

(1) Start with a maximum font size,
(2) Perform greedy word wrapping,
(3) Reduce font size iteratively within bounding box,
(4) Center text within the writable area.

Overflow is detected if the rendered pixels extend beyond the box mask.

3.5 Evaluation Metrics

We evaluate the following:

- Average OCR confidence,
- Empty OCR rate,
- Translation length expansion ratio,
- Text overflow rate,
- Final font size statistics.

Figure 1: Top: Original panel. Middle: OCR bounding boxes after merging and overlap removal. Bottom: Final translated panel with re-typeset English text.

4 Preliminary Results

We evaluated COMET on a development set of 10 manga panels spanning single-bubble scenes, multi-panel layouts, and dialogue-dense pages. Figure 1 illustrates a representative example at three stages: original input, OCR detection overlay after bounding-box merging and overlap removal, and the final translated rendering. Across the dataset, OCR successfully extracted non-empty text from approximately 85% of detected regions. Post-processing through box merging substantially reduced duplicate or fragmented detections leading to more coherent translations and improved layout stability.

In panels with moderate dialogue density, the layout-aware resizing algorithm successfully prevented overflow while maintaining legibility. The iterative font-scaling strategy ensured that translated text remained contained within bounding regions. Most panels preserved visual alignment between speech bubbles and translated text, demonstrating that the pipeline can maintain structural consistency even when text length increases.

Failure cases primarily stemmed from OCR detection challenges. Extremely bold fonts, very small characters, or densely clustered text reduced detection reliability. In such cases, PaddleOCR either failed to detect text entirely or split characters into multiple narrow boxes, producing semantically incomplete translations. These OCR errors propagate downstream to translation and rendering, highlighting that detection robustness remains the primary determinant of end-to-end performance.

5 Discussion and Limitations

The results demonstrate that a modular OCR–translation–layout pipeline can produce visually coherent manga translations without supervised training or manual annotation. Bounding-box merging and overlap removal were particularly important in multi-dialogue panels, where fragmented detections would otherwise result in redundant translations. This refinement stage significantly improved readability and reduced visual clutter in the final renderings.

Despite these strengths, OCR robustness remains the primary bottleneck. Stylized typography, exaggerated stroke thickness, and small-font dialogue frequently degrade detection accuracy. In dense multi-panel layouts, fragmentation or missed detections lead to semantic gaps in translation. Because translation operates directly on OCR outputs, these detection errors propagate through the pipeline and cannot currently be corrected downstream.

Translation quality introduces additional variability. Although the MarianMT model performs reasonably well on standard Japanese text, informal dialogue and context-dependent expressions occasionally produce mistranslations. Since translation is performed independently for each bounding box, the system lacks cross-bubble contextual awareness, which reduces narrative coherence in extended conversations.

Finally, while heuristic font-size reduction effectively prevents overflow in most cases, aggressive downscaling may reduce readability when English expansion substantially exceeds the original text footprint. Additionally, simple whitening-based text removal may introduce minor artifacts in complex bubble interiors. Future work will incorporate evaluation against professionally translated ground-truth panels using automated metrics such as BLEU to more rigorously assess translation accuracy.

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